

Cell-type deconvolution methods for spatial transcriptomics

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Abstract

Spatial transcriptomics is a powerful method for studying the spatial organization of cells, which is a critical feature in the development, function and evolution of multicellular life. However, sequencing-based spatial transcriptomics has not yet achieved cellular-level resolution, so advanced deconvolution methods are needed to infer cell-type contributions at each location in the data. Recent progress has led to diverse tools for cell-type deconvolution that are helping to describe tissue architectures in health and disease. In this Review, we describe the varied types of cell-type deconvolution methods for spatial transcriptomics, contrast their capabilities and summarize them in a web-based, interactive table to enable more efficient method selection.

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Introduction

The spatial organization of cells is fundamental to multicellular life as it influences development, function and evolution. During development, the precise spatial arrangement of cells ensures proper differentiation and the correct formation of tissues and organs, which is crucial for their growth and function^{1,2}. In fully developed organisms, this spatial organization enables efficient cell–cell communication, resource distribution and responses to environmental changes^{3,4}. Pathological conditions often involve disruptions to this spatial organization, such as in some cancers in which uncontrolled cell growth and altered organization compromise tissue function and trigger systemic health issues^{5,6}. As such, insights into spatial organization are vital to understanding basic multicellular biology and mechanisms underlying human health and disease.

Recent advances in RNA-sequencing (RNA-seq) technologies have revolutionized our ability to analyse gene expression at high resolution⁷. More specifically, spatial transcriptomics – of which the first widely adopted technology was described in 2016 (ref. 8) – now enables transcriptional profiling within the native tissue environment⁹. Spatial transcriptomics technologies are either imaging-based or sequencing-based, each with their own benefits and drawbacks. Imaging-based spatial transcriptomics relies on the hybridization and imaging of RNA probes to measure the expression of 200–1,000 genes at single-cell and even subcellular resolution¹⁰. By contrast, sequencing-based spatial transcriptomics measures the expression of all genes but often falls short of single-cell resolution¹⁰. For example, sequencing-based technologies such as Visium v1/v2 (ref. 8), Slide-seq v1 (ref. 11), DBIT-seq¹² and GeoMX DSP¹³ often combine multiple cells within each spatial capture area (also referred to as spots). Newer technologies such as Visium HD offer subcellular resolution but, in practice to date, the fine-grained layout and high dropout level currently require the capture areas to be aggregated to yield a reasonable number of genes. Therefore, analyses of this type of data, which are being rapidly generated by the community, still encounter issues stemming from lacking single-cell resolution.

Owing to the limited resolution of sequencing-based spatial transcriptomics, a plethora of methods have been developed for cell-type deconvolution, which estimates the contribution of specific cell types towards the overall transcriptome measured at each spatial location. These methods typically output proportions of cell types at each spatial location. With numerous deconvolution methods available to date, choosing the most effective approach depends on many factors related to the technology used, experimental setup and study objectives (Supplementary methods).

In this Review, we provide an overview of methods for cell-type deconvolution of sequencing-based spatial transcriptomics (hereafter referred to simply as spatial transcriptomics, unless specified otherwise). We first explain in more detail the inputs and outputs of cell-type deconvolution, highlighting the distinction between methods that do and do not use a reference dataset. Next, we categorize

tools based on their deconvolution frameworks and then describe how they incorporate different modalities such as single-cell data, spatial coordinates and haematoxylin and eosin (H&E)-stained images. Of note, we focus on cell-type deconvolution but not on methods devoted to related yet distinct tasks such as spatial and single-cell data integration (for example, Harmony¹⁴ and LIGER¹⁵), which are primarily aimed at addressing batch effect rather than estimating cell-type contributions. Finally, we offer a [web-based summary table](#) of all reviewed cell-type deconvolution methods with details on several characteristics, such as the data preprocessing steps, capacity for additional modalities, datasets used for validation, benchmarking status and documentation link (Supplementary Tables 1–4). The summary table also has interactive filtering that enables the user to narrow down methods based on specific characteristics, and researchers can submit new methods through an [update form](#) to keep the summary table current.

Deconvolution inputs and outputs

Spatial transcriptomics captures and sequences RNA from specific regions of a tissue section at a spatial resolution defined by capture areas of varied size and spacing (for instance, 10x Visium has 55 μm diameter and 100 μm centre-to-centre spots⁸; Slide-seq has 10 μm diameter and 10 μm centre-to-centre beads¹¹). The first preprocessing step of spatial transcriptomics – similar to standard transcriptomics – is to produce a count matrix of gene expression values. Next, cell-type deconvolution is applied to the count matrix, optionally with the addition of annotated single-cell RNA-seq (scRNA-seq) data as a reference (Fig. 1a,b). This approach to deconvolution has been extensively used on bulk RNA-seq data^{16,17}; however, spatial transcriptomics has a lower sequencing depth and captures smaller cell populations (in each spot) than bulk RNA-seq, such that specialized deconvolution methods are required for spatial transcriptomics with or without the use of a reference dataset. Moreover, cell-type deconvolution does not solely entail integrating spatial with scRNA-seq data, but rather leverages as many data types as possible, such as histological images, for better cellular profiling.

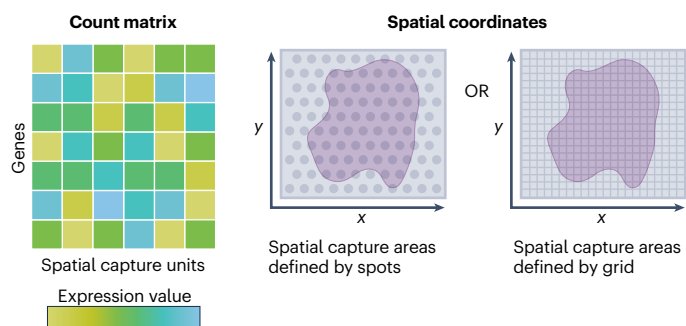
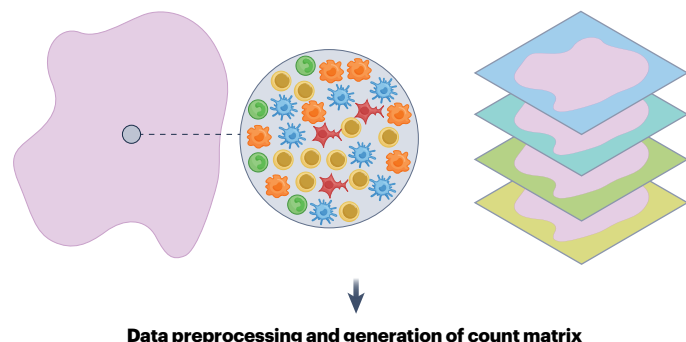
Reference-based deconvolution

Deconvolution methods can be classified based on whether they use annotated scRNA-seq data as a reference for the gene expression of individual cells. Reference-based methods use scRNA-seq data by modelling spot gene expression as a mixture of single-cell gene expression or by building representative cell-type gene expression profiles¹⁸. These methods benefit from external reference datasets and therefore provide results coherent with prior knowledge of tissue composition deciphered through single-cell studies¹⁹. However, these methods can propagate inaccuracies from the scRNA-seq annotation to the spatial transcriptomics analysis. Furthermore, they do not enable detection of novelties, such as uncharacterized cell states only captured in spatial transcriptomics (if they exist), and the scRNA-seq data might

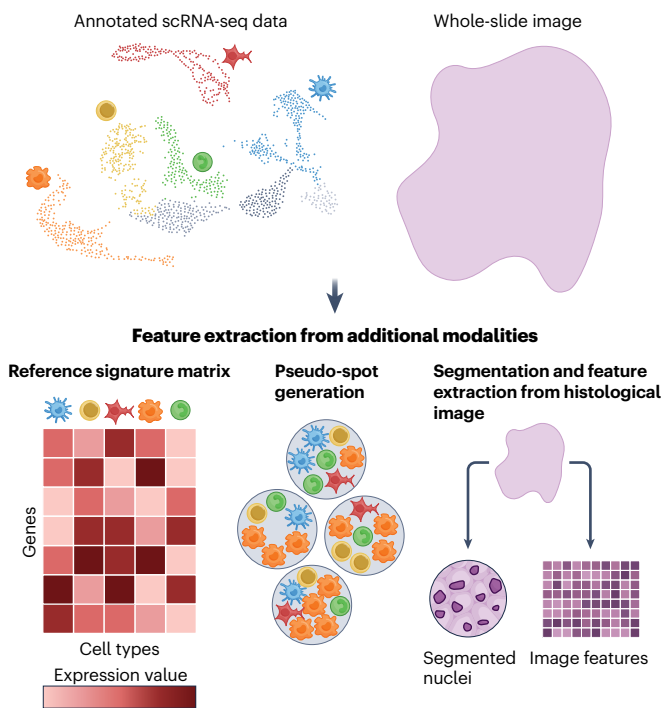
Fig. 1 | Principal workflow of cell-type deconvolution for spatial transcriptomics. **a**, Spatial transcriptomics data are collected with sequencing-based technology from a tissue of interest. Data are preprocessed with a pipeline for quality control and alignment to reference genome. **b**, Concomitantly, additional modalities such as annotated single-cell RNA-sequencing (scRNA-seq) data and image with staining can be generated to help deconvolution. These datasets are transformed: scRNA-seq data can be used to build a reference signature matrix or generate pseudo-spots, and the image can be

segmented or subjected to a feature extractor. **c**, Preprocessed data from spatial transcriptomics and other modalities are used as input to the deconvolution algorithm, which returns the contribution of each cell type at each location in different forms, including proportion or count matrix, mapping matrix, cell locations and reconstructed single-cell gene expression profile. A proportion matrix is the most common output. **d**, Downstream analysis and visualization of the deconvolution results enable the user to gain insights into cell-type distribution and spatial organization.

a Sample collection for spatial transcriptomics analysis

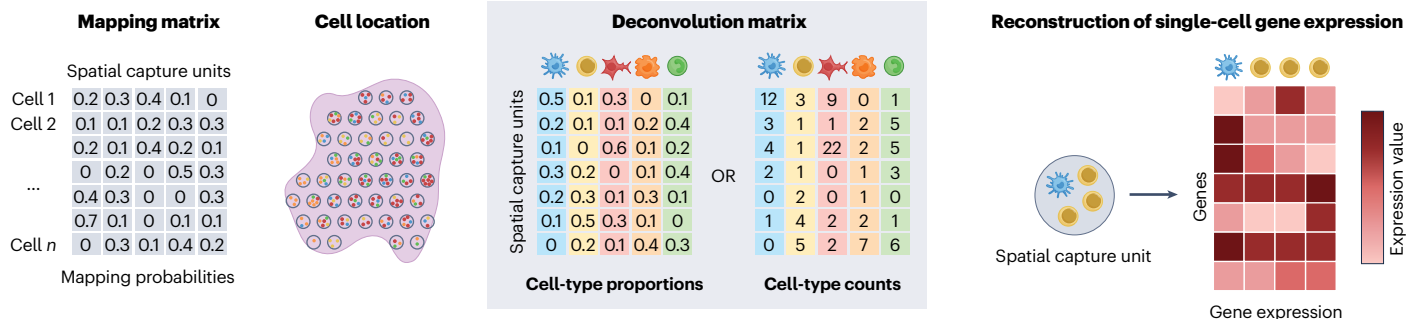


b Extra-modality integration

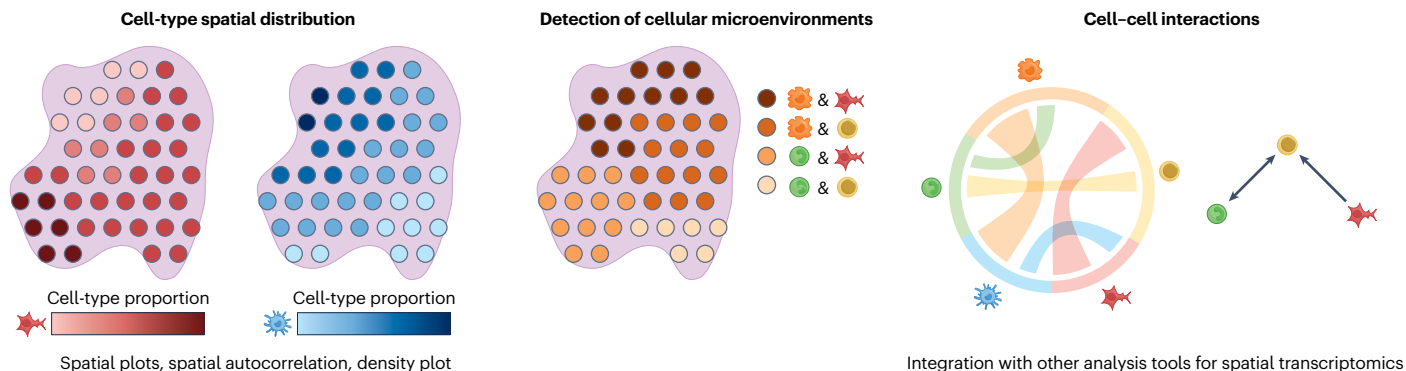


Deconvolution

c Possible outputs



d Downstream analysis and visualization



not always be fully representative of the cell-type coverage of spatial transcriptomics data, as assumed by many of these methods²⁰.

Reference-free deconvolution

Reference-free methods do not use scRNA-seq data for the deconvolution itself and therefore estimate cell composition without previously defined cell-type expression profiles. Instead, these methods use factor decomposition to analyse any correlations and intrinsic variation in spatial transcriptomics data that reveal the transcriptomic contributions of different cell types. In contrast to reference-based deconvolution methods, reference-free methods import prior knowledge downstream of the decomposition task when analysing the inferred cell types, which avoids introducing biases from scRNA-seq data (owing to technological differences, for example). Reference-free methods are particularly useful when studying tissues that have not been characterized at the single-cell level or when uncovering new cell states. However, the central assumption of these methods is that cell types are represented by ‘factors’, which are gene programmes whose annotation is highly dependent on the chosen database and may also reflect other biological processes. Moreover, inferring cell type without a reference is less constrained, and thus harder to solve, and the obtained solution might not align well with prior knowledge.

Outputs

Deconvolution methods produce diverse outputs, each offering unique insights into cell composition and gene expression in tissues (Fig. 1c). The main outputs include:

- **Proportion per cell type:** the relative abundance of each cell type contributing to the observed gene expression for each spot. Some methods also provide probabilities of the presence of each cell type in a spot, often interpreted as proportions. In a reference-free context, the output is the proportion of each factor, which are further annotated as cell types.
- **Cell counts:** the absolute number of cells of each type in a spot. This approach is more precise and informative than proportions, as it gives a direct measure of cell density and avoids the biases associated with probabilistic or proportional interpretations, particularly when the total number of cells is low.
- **Gene expression profile:** the levels of gene expression for each cell type, which are rarely obtained by deconvolution methods but are extremely valuable for providing an in situ transcriptome of the cell type and describing potential cell states only found in spatial transcriptomics.
- **Individual cell mapping:** a matrix in which each column is a spot from the spatial transcriptomics data and each row is a cell from the scRNA-seq data. As such, this output is produced only by reference-based methods that use external scRNA-seq to spatially map individual cells to spatial transcriptomics spots.
- **Cell location:** an estimation of the precise location of individual cells present in each spot, often visualized on the histological image, in addition to the cell type or gene expression information.

These deconvolution outputs are further integrated into the spatial transcriptomics data analysis pipeline to yield insights into tissue architecture and functions (Fig. 1d).

Deconvolution frameworks

Cell-type deconvolution tools use advanced statistical methods that can be categorized by their mathematical frameworks, which are

presented in increasing complexity order and include regression, dimensionality reduction, Bayesian, optimal transport and deep learning techniques (Fig. 2). Each framework has unique characteristics and limitations, and the most suitable approach depends on the specific dataset and research goals (Fig. 3, Supplementary Tables 1–4 and [web-based summary table](#)).

Regression models

Deconvolution with regression models entails reconstructing the spatial transcriptomics data from known gene expression profiles (that are typically inferred from scRNA-seq data) and then interpreting the coefficients of the regression model as the cell-type proportions. Two types of regression approaches, non-negative least squares (NNLS) and dampened weighted least squares (DWLS)^{16,21–24}, have been developed for deconvolution of bulk RNA-seq and since been adapted for deconvolution of spatial transcriptomics data. For example, SpatialDWLS²⁵ builds on DWLS by performing an enrichment analysis on spatial data, whereas Cottrazm²⁶, a method designed specifically for the cancer microenvironment, performs two rounds of DWLS. Additionally, several cell-type deconvolution methods use NNLS (with unique adaptations), owing to its high interpretability and suitability for deconvolution by ensuring non-negative cell-type contributions. For example, AdRoit²⁷, NLSDeconv²⁸ and Redeconve²⁹ all apply regression models directly to scRNA-seq, but Redeconve introduces a regularization term to address gene expression collinearity among cells of the same type whereas NLSDeconv focuses on computational efficiency. Another tool, SPOTlight³⁰, integrates NNLS with dimensionality reduction and yet another, SpaCET³¹, tailors NNLS specifically for deconvolution of the cancer microenvironment.

In contrast to NNLS and DWLS linear regression models, nonlinear regression models offer greater flexibility in capturing the complexity of cell-type contributions at each spatial location. For instance, CellTrek³² trains a multivariate random forest model to measure the similarity between spatial transcriptomics and scRNA-seq data. SpatialPrompt³³ uses a *k*-nearest neighbours regressor to approximate spot composition by relying on simulated data. SPADE³⁴ introduces Lasso-regularized regression to estimate the number of cell types in each spatial domain. SpatialDecon³⁵ implements log-normal regression to correct for noise from technical variation. Finally, SONAR³⁶ uses a geographically weighted regression to explicitly model spatial heterogeneity and account for location-dependent variation in gene expression.

All these regression frameworks are interpretable and straightforward to implement. However, caution should be taken when using these tools for spatial deconvolution because regression models may be affected by correlations or multi-collinearities often observed in gene expression and across spatial locations³⁷.

Dimensionality reduction

Dimensionality reduction techniques uncover hidden structures in gene expression data by decomposing the expression matrices into ‘factor by spot’ and ‘gene by factor’ matrices. The first (factor by spot) matrix represents the cell-type presence in each spot and the second (gene by factor) matrix captures gene signatures, which are characteristic gene expression patterns associated with each cell type. Given the non-negative nature of the cell-type proportions or counts in these matrices, several dimension reduction methods for spatial cell-type deconvolution use non-negative matrix factorization (NMF). For instance, CARD³⁸ uses a spatial autocorrelation model on top of the

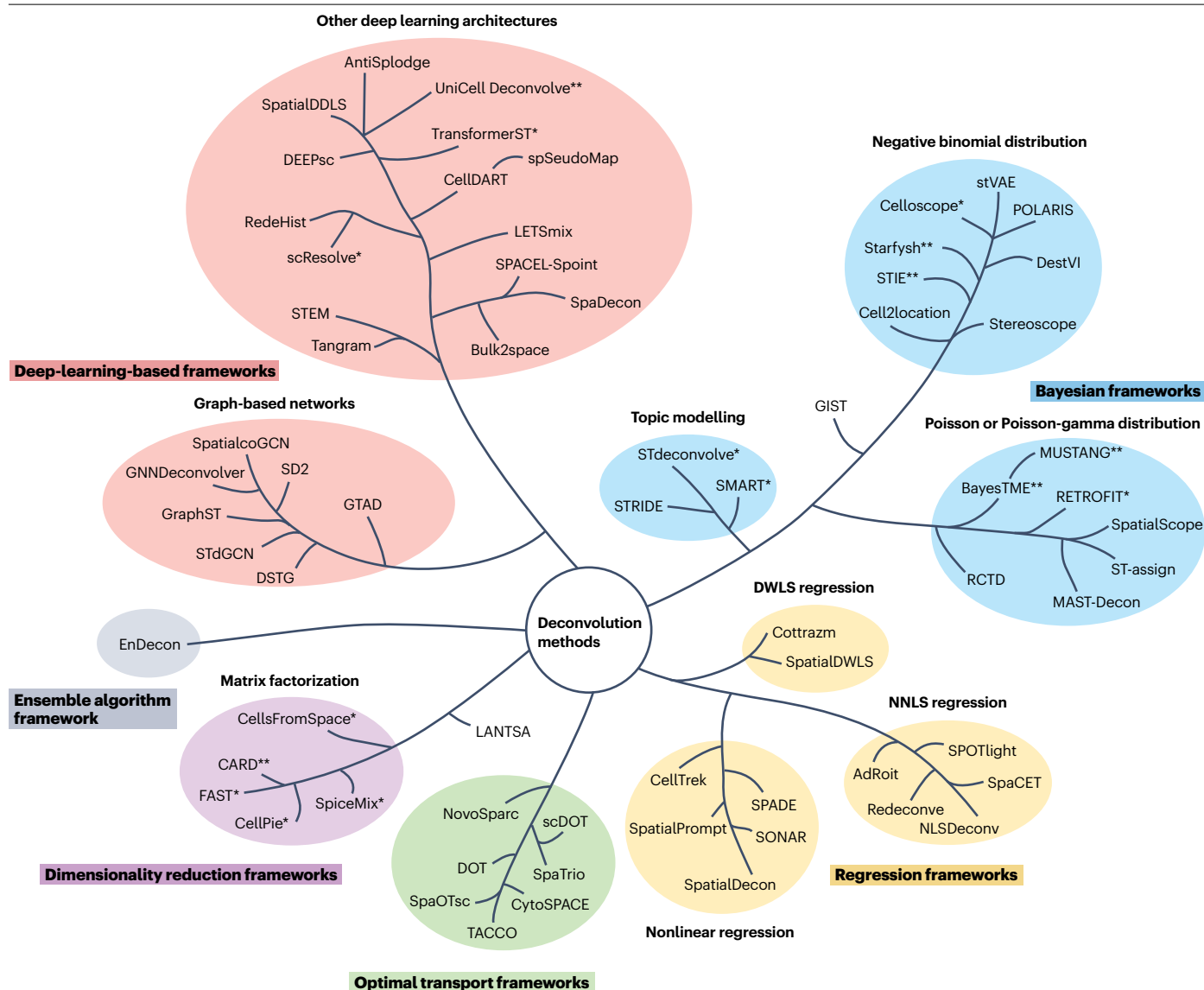


Fig. 2 | Classification of cell-type deconvolution methods for spatial transcriptomics. Methods are grouped by the frameworks used to perform deconvolution. Bayesian models are divided into three subtypes (blue). The first is topic modelling and the other two correspond to two distribution assumptions used in classical Bayesian frameworks – negative binomial and Poisson (or Poisson-gamma) distributions. GIST has its own branch, as it uses a t -distribution. Regression models also fall into several categories, differing from one another by, for example, data assumptions and imposed constraints (yellow). Optimal-transport-based methods are grouped into a single category (green), as are dimensionality reduction methods, which are mainly based

on matrix factorization (purple), except for LANTSA (landmark-based). EnDecon stands alone in its category as the only consensus method among the other deconvolution methods (grey). Finally, most methods belong to the deep-learning-based category, with a subcategory dedicated to graph-based networks (red). The other deep learning architectures use autoencoders, fully connected neural networks or others to perform deconvolution. *Reference-free methods. **Both reference-free and reference-based implementations; methods that can be used with or without a reference. DWLS, dampened weighted least squares; NNLS, non-negative least squares.

NMF; SpiceMix³⁹ leverages probabilistic graphs to incorporate the spatial organization of cells; and both CellPie⁴⁰ and FAST⁴¹ incorporate spatial data and histology, although FAST uses a regularized version of NMF compared with the joint factorization of CellPie. Beyond NMF, independent component analysis, which was already proposed for bulk RNA-seq deconvolution^{42,43}, has also been applied to spatial transcriptomics deconvolution. In contrast to NMF, independent component

analysis assumes statistical independence of components without the non-negative constraint, as implemented in CellsFromSpace⁴⁴. Other innovative approaches are currently being developed, such as LANTSA⁴⁵, in which spatial data representation is learned from a smaller subset of samples called ‘landmarks’.

Dimensionality reduction methods are particularly valuable for deconvolution when no scRNA-seq data are available or if the aim is to

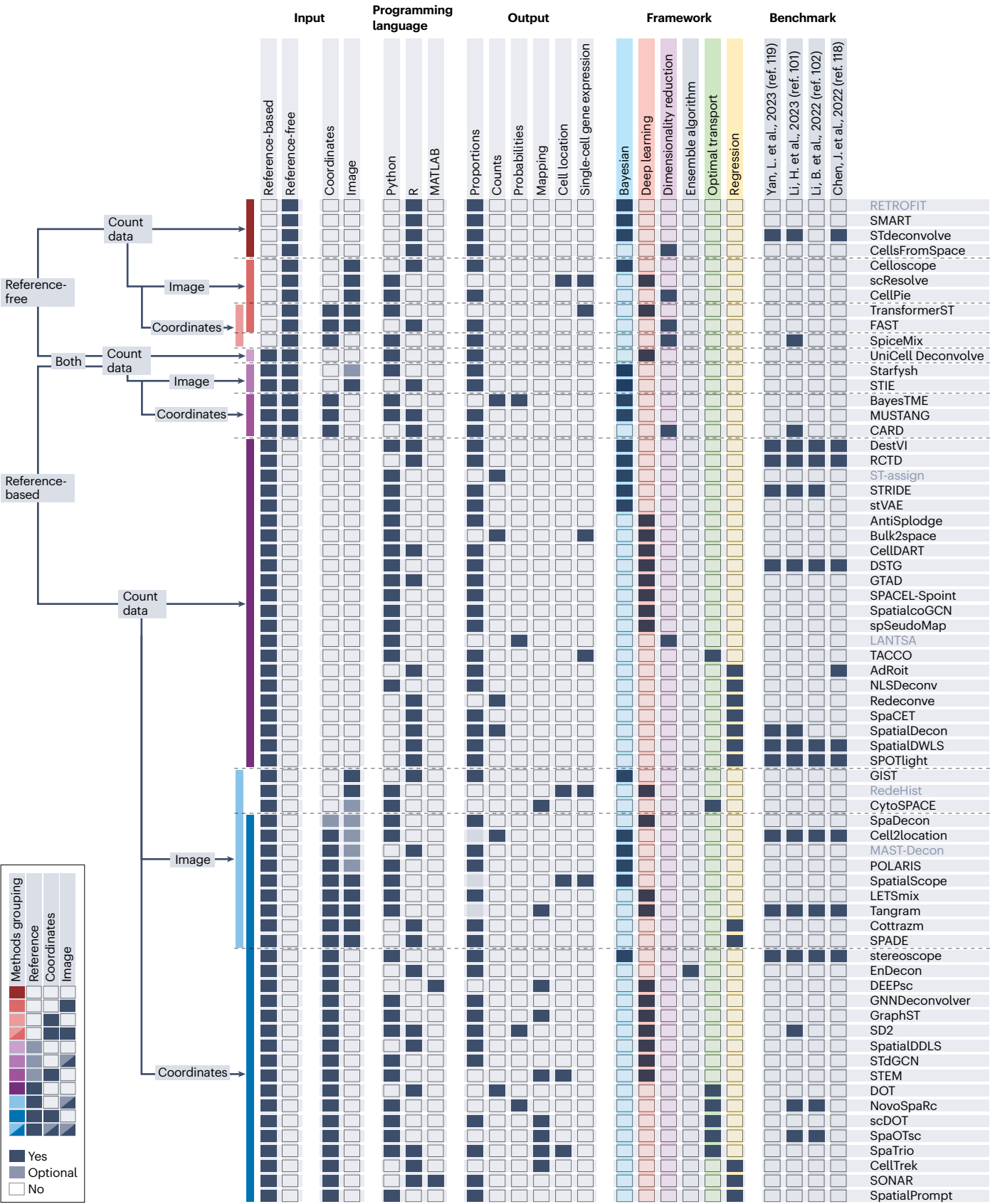


Fig. 3 | Characteristics of deconvolution methods for spatial transcriptomics.

Summary of deconvolution method characteristics across six categories: use of reference, use of extra modality (coordinates and/or image), programming languages according to GitHub repositories, output types, class of framework following our proposed classification and presence in independent

benchmarks (Supplementary Fig. 2). Methods are grouped according to shared characteristics. Method names in grey are methods not yet peer reviewed.

Left: decision tree to identify all possible methods in function of the data types available and the modalities the user wants to use. The benchmark studies cited in this Review are refs. [101,102,118,119](#).

uncover new cell types or states. However, the assumption that the factors revealed by dimensionality reduction correspond to distinct cell types is not always valid, as they may instead represent broader biological processes, technical artefacts or mixed signals. Furthermore, annotating factors relies on gene set enrichment and knowledge databases that need to be selected carefully. Although these methods are highly interpretable and computationally efficient, they face additional challenges related to initialization sensitivity and difficulty in handling spatial data that are noisy or contain closely related cell types.

Bayesian models

Bayesian methods for deconvolution estimate cell-type contributions by assigning probability distributions to any contributing variable – such as gene, cell-type specific signatures, image features, technical effects and other unobserved variables – and then modelling their dependencies. These methods commonly use gene signatures or marker gene lists (that are often derived from scRNA-seq data) to guide the estimation process. Once the probabilistic rules, such as prior distributions and conditional dependencies between variables, have been determined, the model trains itself to fit the different distributions and estimate various parameters, including cell-type contribution, batch effects and platform effects.

When using Bayesian models for deconvolution of spatial transcriptomics data, probability distributions are often modelled using negative binomial^{46–53} and Poisson or Poisson-gamma distributions^{54–60} – only GIST⁶¹ implements a *t*-distribution. However, these statistical models vary in how they learn the reference matrix, model cell-type contributions, consider spatial location and account for technology differences. For instance, Starfish⁵³ uses archetypal analysis to identify spots likely containing a single cell type and RCTD⁵⁴ proposes a corrective term to adjust for technology differences. Similarly, Cell2location⁴⁸ leverages a parametrized negative binomial distribution with cell-type contribution in the negative binomial rate, whereas POLARIS⁵¹ introduces a layer-specific parameter that enables the model to take into account the spatial organization of tissue layers.

Bayesian topic modelling approaches, originally developed for natural language processing, have been adapted for deconvolution. These models identify topics which are gene programmes and are under the key assumption that they represent cell types. Consequently, the correspondence between topics and cell types largely depends on the signature databases used, which may contain both highly specific cell-type signatures and broader, more general ones like pathways^{62–64}. Two Bayesian topic modelling tools, STdeconvolve⁶² and STRIDE⁶³, implement latent Dirichlet allocation generative models⁶⁵ – STdeconvolve uses it directly on the spatial transcriptomics data, whereas STRIDE leverages topic profiles trained from scRNA-seq. Conversely, SMART⁶⁴ implements the keyATM⁶⁶ model, which incorporates predefined genes associated with topics to influence the model learning process.

Bayesian models for deconvolution benefit from being able to incorporate prior knowledge, provide full probabilistic interpretability and quantify uncertainty of the results. However, these approaches

require various inference strategies to compute maximum a posteriori or maximum likelihood estimators, such as Monte Carlo algorithms or variational inference, which are often computationally intensive and slow when GPU acceleration is not provided. Additionally, selecting appropriate prior distributions and hyperparameters is difficult, as it involves subjective choices that can influence model stability, convergence and the reliability of inferred results.

Optimal transport

Optimal transport methods apply mathematical optimization to align and compare gene expression data from scRNA-seq and spatial transcriptomics. These approaches match gene expression patterns between the two datasets as efficiently as possible by minimizing the effort (or ‘cost’) required to transform the distribution of gene expression value from scRNA-seq into that of the spatial transcriptomics. In doing so, deconvolution with optimal transport maps cell-type signatures or individual cells from scRNA-seq data to their spatial locations in the tissue while minimizing discrepancies between the two data sources. This framework is used by several deconvolution methods but with distinct problem formulations and cost functions^{67–73}. For example, the pioneering tool, NovoSparc⁶⁷, was initially developed to infer spatial distributions directly from scRNA-seq data. Two tools that came later, SpaTrio⁶⁸ and scDOT⁷⁰, integrate optimal transport in part by defining the cost function with transcriptomics dissimilarity (that is, a measure of how different gene expression profiles are between datasets) combined with NNLS regression. TACCO⁶⁹ introduces yet another unique formulation of optimal transport for deconvolution – semi-unbalanced entropic optimal transport with a different similarity metric, which allows higher flexibility in distribution matching. Another tool, DOT⁷¹, leverages the local spatial context to address gene expression differences between scRNA-seq and spatial transcriptomics.

Optimal transport is a highly versatile, flexible and powerful framework for comparing scRNA-seq and spatial transcriptomics data distributions with the goal of cell-type deconvolution. However, its effectiveness heavily depends on the choice of cost function and parameter settings. Additionally, these methods can suffer from incomplete feature overlap between scRNA-seq and spatial transcriptomics data, which can impede efforts to incorporate any differences between data generation protocols. Finally, as datasets become larger, optimal transport methods may face challenges with computational scalability if specific strategies (such as those used in TACCO⁶⁹) are not implemented across tools.

Deep learning models

Tools that use deep learning models approach cell-type deconvolution by harnessing neural networks to identify and learn complex patterns in spatial gene expression data. These models are particularly useful for large, high-dimensional datasets and thus represent the major framework for cell-type deconvolution from spatial transcriptomics^{74–95}. Typically, these models include feature extractors that can process spatial transcriptomics data, and in some cases histology images, to identify key features related to tissue structure. Various models are

Box 1 | Pseudo-spot generation

Simulated spatial transcriptomics data are used by most deconvolution methods for model validation. In addition to validation, some methods generate pseudo-spots for model training or to address challenges such as data scarcity and imbalances in cell-type proportions. For instance, LETSmix⁸⁸ generates pseudo-spots to train cell-type deconvolution and a real-spot versus pseudo-spot classifier in an adversarial manner, thus enhancing model robustness. Moreover, UCDBase⁷⁹ generates millions of pseudo-bulk RNA mixtures to improve deconvolution model performance, particularly for rare cell types. Therefore, pseudo-spot generation and associated strategies are a central aspect of deconvolution.

Pseudo-spots can be completely synthetic if built from simulated single-cell data, or they can be semi-synthetic if built from a real single-cell RNA-sequencing (scRNA-seq) dataset. Additionally, sampling strategies for pseudo-spot generation from scRNA-seq data can be random or based on known distributions. The sampling strategy must also be adapted to represent spatial transcriptomics data, for example, by controlling for unbalanced cell-type proportions between scRNA-seq and spatial transcriptomics data. To ensure that scRNA-seq data are transformed into a realistic spatial transcriptomics dataset, one can also adjust the number of cell types per location^{50,79,92}, pick distributions to ensure cell-type representation and varied pseudo-spot composition^{75,93,95}, downsample UMI (unique molecular identifier) counts if they exceed true spatial data⁹² or balance the pseudo-spot quantity for optimal performance⁸⁷. Ultimately, the gene counts from the selected cells are aggregated to represent spatial data as the weighted contribution of different cell types. By contrast, pseudo-spot generation from imaging-based spatial transcriptomics data is achieved by binning single cells and is mostly used for validation purposes. Overall, the effectiveness of pseudo-spots relies not only on the assumptions and algorithms but also on the quality of scRNA-seq annotation. Other limitations of pseudo-spots include not fully capturing the complex spatial relationships and biological variability as well as inconsistencies owing to differences in experimental conditions, such as sequencing depth and batch effects. These situations inevitably lead to poor model performance or misleading model evaluations.

trained on pseudo-spots built from scRNA-seq (Box 1), sometimes in combination with gene signatures. Once trained, the model is used on real spatial transcriptomics data to predict cell-type contribution at each spot. The diversity of deconvolution tools that use deep learning models is driven by the various deep learning architectures, which enables flexibility and adaptability in handling different data types. For example, SPACEL⁷⁶, SpatialcoGCN⁹⁰ and Bulk2space⁷⁴ use a variational autoencoder for gene expression features, whereas DEEPsc⁷⁸ and TransformerST⁸³ leverage fully connected neural network and vision transformers, respectively, to integrate histology images. Beyond these three types of architectures, other deconvolution tools use foundational models⁷⁹, attention networks⁸⁷, multilayer perceptron^{76,84} and graph neural networks^{90–95} (GNNs), illustrating the large variety of techniques. GNNs are frequently used for deconvolution owing to their ability to handle spatial transcriptomics data represented as a graph, in which

nodes represent spots and edges represent the spatial distance or gene expression similarity between spots⁹⁶. GNNs then perform deconvolution by integrating pseudo-spots with known cell-type contributions in the graph. Different GNN-based methods use varied graph specifications and learning strategies that are tailored to answering deconvolution needs.

Methods that use deep learning can capture complex and intricate relationships from multiple modalities and identify subtle patterns, which is particularly useful for learning gene expression patterns and image classification. These methods can make highly accurate predictions but also suffer from the ‘black box’ nature of deep learning that hinders the ability to understand or interpret how the model arrives at these predictions, making understanding of model failures difficult. As such, future deconvolution methods may focus on explainability to inform how these deep learning models are handling spatial and multimodal data.

Multimodal data integration

Cell-type deconvolution of spatial transcriptomics data can be improved by incorporating data from other modalities collected externally or in parallel. For example, to determine cell-type distributions in spatial transcriptomics data, most deconvolution methods rely on the cell-type-specific expression profiles commonly available from scRNA-seq studies and community-curated atlases^{97–99}. Additionally, information on the spatial localization of transcripts and the H&E-stained tissue images collected in parallel to the gene expression can offer valuable and complementary information for deconvolution. A set of methods have been designed to leverage these multimodal datasets for cell-type deconvolution of spatial transcriptomics data (Figs. 3 and 4), although users should carefully consider the most suitable method based on data availability and quality, tissue characteristics and their biological questions (Box 2).

Integrating single-cell RNA-sequencing

Deconvolution of spatial transcriptomics facilitated by the integration of scRNA-seq data are reference-based analysis frameworks that shift data interpretation from unsupervised to supervised. Deconvolution with prior knowledge from one or several external annotated scRNA-seq datasets entails either building a reference gene signature matrix, building pseudo-spots corresponding to simulated spatial transcriptomics spots with known cell-type composition, or directly mapping the cells from the scRNA-seq onto the spatial transcriptomics data (Fig. 4a).

Building a reference gene signature matrix. Reference gene signature matrices store the characteristic gene expression values of cell types present in scRNA-seq data. These matrices improve deconvolution by helping to distinguish similar cell types, standardize comparisons across samples and obtain deconvolution results compatible with prior knowledge. Before building the reference matrix, each method typically involves preprocessing steps common to scRNA-seq analysis, including gene and cell filtering, normalization and then gene name intersection with spatial data. Such quality control steps are important because the reference matrix quality often has a greater effect on the result than the deconvolution method itself, as demonstrated on bulk RNA-seq deconvolution¹⁰⁰.

Benchmarking studies have shown that cell-type deconvolution methods perform optimally when following the recommended steps for normalization and gene selection during reference matrix

construction^{101,102}. In terms of gene selection, some methods (for example, GraphST⁹¹ and DestVI⁴⁷) focus on highly variable genes to capture the broad pattern of the scRNA-seq, whereas others (for example, CARD³⁸ and RCTD⁵⁴) perform differential gene expression analyses to identify key marker genes for each cell type. Next, methods typically average the gene expression of all cells of the same type or use regression approaches to obtain a gene signature⁴⁸. Importantly, gene signatures must be carefully validated and continuously refined to address deconvolution challenges arising from dynamic cellular states and technical challenges related to data quality annotation and integration.

Building pseudo-spots in silico. Pseudo-spots are synthetic spatial transcriptomics capture areas generated from scRNA-seq or spatial transcriptomics (mainly imaging-based) data at single-cell resolution. The pseudo-spots are simulated with a sampling strategy that draws single-cell information from the original dataset incorporating or not spatial relationships. Then, gene expression profiles of individual cells are aggregated within each pseudo-spot (Box 1). As the data are simulated, the cell types and numbers are known, providing researchers data with a ground truth, which helps validate but also train and optimize deconvolution methods. For example, SpatialDDLs⁸⁵ generates pseudo-spots to train a neural network capable of estimating cell-type proportions from new spatial transcriptomics data. Pseudo-spots can also be used to artificially increase data size and completeness, which facilitates integrating multiple spatial transcriptomics datasets, model training and label transfer from known annotations. However, pseudo-spots can be limited in their ability to properly model spatial transcriptomics data owing to the complex spatial correlations and biological variability of such datasets (Box 1).

Directly mapping single cells. Some cell-type deconvolution methods directly map single cells from scRNA-seq data to spatial transcriptomics capture areas. This approach provides high-resolution visualization of single-cell localization and gene expression but can sometimes result in incomplete spatial coverage. Methods such as STEM⁸⁴, Seurat¹⁰³, CellTrek³², SpatialScope⁶⁰, RedeHist⁸¹, Tangram⁷⁷, GraphST⁹¹, SpaTrio⁶⁸ and scDOT⁷⁰ use this strategy, producing matrices of mapping probabilities or spatial single-cell gene expression profiles instead of proportions. STEM⁸⁴, Seurat¹⁰³ and CellTrek³² use co-embedding of scRNA-seq and spatial transcriptomics data, with Seurat¹⁰³ and CellTrek³² applying canonical correlation analysis and transfer learning to assign cells to locations. scDOT⁷⁰ uses optimal transport techniques to minimize distances or dissimilarities between scRNA-seq and spots from spatial transcriptomics. GraphST⁹¹ uses contrastive learning to learn mapping matrices separately for spatial areas and cell representations. However, mapping methods can suffer from sparse spatial coverage depending on model specification and, particularly for optimal transport algorithms, sufficient overlap of genes captured by scRNA-seq and spatial transcriptomics data is required to not overly simplify the spatial complexity. These methods can also struggle with accurately mapping cells in regions with lower expression levels or complex cell-type compositions.

Controlling for technology effects. Using scRNA-seq to aid cell-type deconvolution from spatial transcriptomics can introduce biases owing to differences between the technologies. These ‘technology effects’ can result from differences in data distribution, dropout rates¹⁰⁴, technical variability¹⁰⁵, cell-type capture efficiency, ambient RNA contamination¹⁰⁶, detection sensitivity, library size and sequencing

depth¹⁰⁷. However, these inherent differences are only addressed by a few deconvolution methods, most of which are Bayesian and regression frameworks that can explicitly account for these biases with corrective variables. For example, Cell2location⁴⁸ and Stereoscope⁴⁶ use scaling parameters in their models, whereas RCTD⁵⁴ and TACCO⁶⁹ correct the gene signature with a parameter inferred through pseudo-bulk analysis of spatial transcriptomics, a process called platform or technology normalization. Other methods account for technologies differences by relying on a shared latent space where spatial transcriptomics and scRNA-seq are jointly represented to perform label transfer across the platforms (for example, in LANSTA⁴⁵, CellTrek³² and TACCO⁶⁹) or use non-parametric statistics (for example, in Bulk2space⁷⁴). These methods are flexible by not assuming a rigid relationship between spatial transcriptomics and scRNA-seq technologies.

Technology effects commonly result from certain cell types not being captured in scRNA-seq, thus preventing their detection with deconvolution. Indeed, cell types missing from scRNA-seq affects deconvolution of bulk RNA-seq, ultimately leading to missing cell-type gene expression being attributed to closely correlated but different cell type^{17,108}. Therefore, all relevant cell types should ideally be represented in the reference data, potentially by combining multiple scRNA-seq datasets to match the expected cell-type diversity in spatial transcriptomics data. When comprehensive scRNA-seq data are not available, several deconvolution methods have implemented solutions to take into account missing cell types. For example, spSeudoMap⁸⁰ offers the possibility to rely on FACS-sorted scRNA-seq data, which is typically selecting or over-representing some cell types, whereas Celloscope⁴⁹ and SpaCET³¹ introduce ‘dummy cell types’. Importantly, these methods allow the discovery of uncharacterized or missing cell states and enable a more flexible cell-type deconvolution.

Another major technology effect can result from differences in dropout and ambient RNA contamination, which are addressed with specialized corrections in scRNA-seq but not generally in spatial transcriptomics data analysis. In the case of deconvolution, these two problems are mostly dealt within optimal transport frameworks by using similarity metrics, such as cosine similarity – a scale-invariant metric that focuses on the direction of the expression profile vector rather than the actual counts. This metric enables effective comparisons between spots and individual cells as vector direction is not expected to change drastically owing to differences in measurement sensitivities between the two technologies, even if the overall expression magnitude of the spot is different from that of the individual cell^{70,71}. Furthermore, one tool named DOT addresses dropout between scRNA-seq and spatial data by leveraging both the shared and unique genes of each modality to perform deconvolution, which enhances the integration of heterogeneous transcriptomic datasets⁷¹.

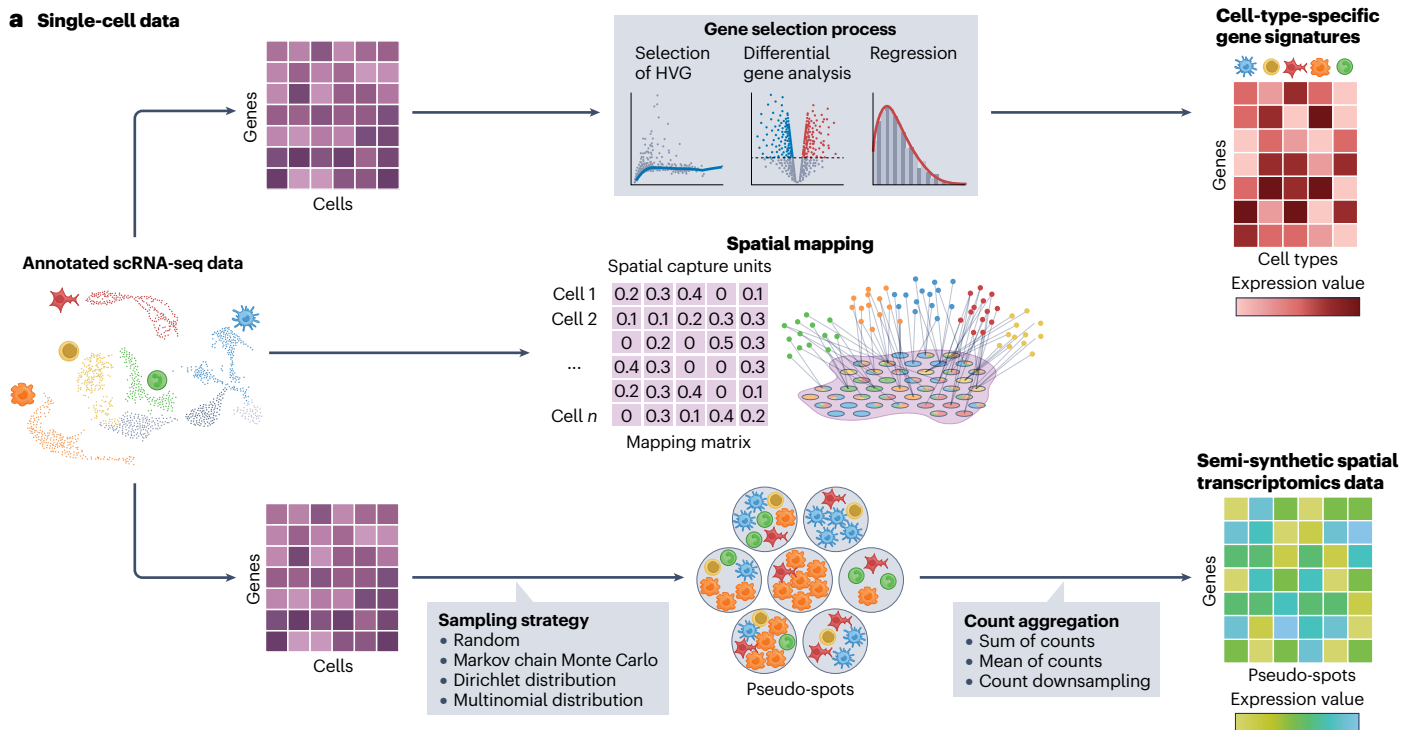
Despite these advances, addressing technology effects in spatial transcriptomics remain an ongoing challenge. As such, continued development is needed to improve bias correction, handle uncharacterized cell types and better adapt methodologies for the unique properties of spatial transcriptomics.

Image-informed deconvolution

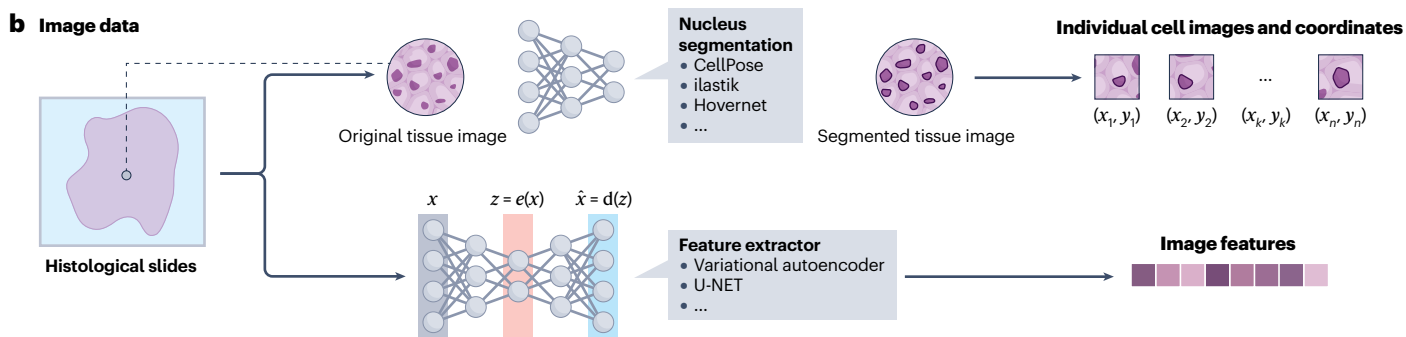
With advancements in spatial transcriptomics technologies, particularly for those that include staining protocols (such as Visium), tissue images are increasingly integrated into deconvolution workflows (Fig. 4b). Histological images offer valuable spatial and phenotypic information that can improve deconvolution by providing information on cell morphology and distribution within spots.

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a Single-cell data



b Image data



c Coordinate data

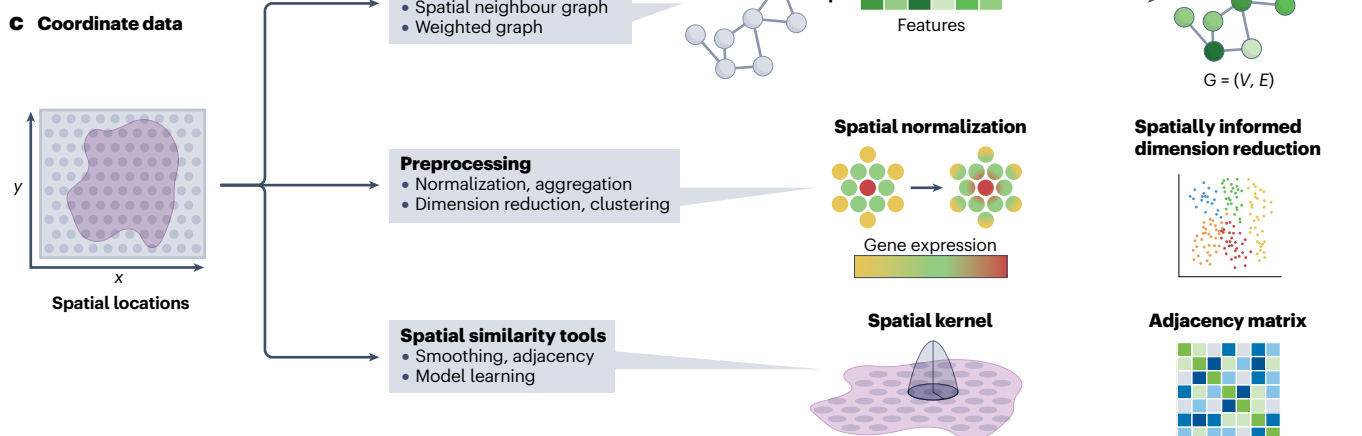


Fig. 4 | Multimodal integration for spatial transcriptomics deconvolution.

Each panel depicts a modality complementing spatial transcriptomics data and how this modality can be processed to be integrated into deconvolution frameworks. **a**, Annotated single-cell data can be used to build a cell-type-specific gene signature matrix (top), directly for mapping (centre) or to generate pseudo-spots (bottom). Integration of single-cell RNA-sequencing (scRNA-seq) data depends on method or user choice. For the construction of a reference matrix, several preprocessing and gene selection steps can be applied upstream. For example, Cell2location⁴⁸ uses binomial regression to remove batch and technology effects. For the generation of pseudo-spots, several sampling methods can be used to distribute cells in synthetic spots, and several types of aggregation can be used to construct pseudo-spatial transcriptomics

data (Box 1). **b**, Histological images are also an accessible and very useful input for spatial transcriptomics data analysis. Many methods use segmentation tools to count cells in spots and determine their spatial coordinates. Several deep learning algorithms are also used to extract morphological features that may be spot-specific or cell-specific. Such features can then be used in the deconvolution methods. **c**, Finally, many approaches are used to consider the spatial coordinates. The spatial organization of capture areas can be considered in preprocessing stages but can also be used as additional data during training, particularly in graph convolutional networks. Some methods (for example, SpatialDDL⁸⁵) use the spatial coordinates after deconvolution to perform a spatial regularization of proportions. HVG, highly variable genes.

Several deconvolution methods incorporate tissue images into their models, including those that use segmentation masks often retrieved with external algorithms such as CellPose¹⁰⁹ and ilastik¹¹⁰ (Table 1 and Fig. 3). This preprocessing step enables the true cell locations and counts to be incorporated into the model and thus tailor parameters based on tissue cellularity. Additionally, some methods leverage deep learning tools developed for image analysis to extract meaningful patterns¹¹¹. For instance, RedeHist⁸¹ applies U-NET¹¹² on image patches to extract pixel-level histological features, and Starfrysh⁵³ integrates variational autoencoders to extract spot-level features from low-resolution images. Importantly, the image quality has a large impact on the quality of the results regardless of how image is used, but is often suboptimal because, until recently, images were used only for visualization. Therefore, future work should generate (and make available) images with optimal resolution and at maximum magnification to ensure reliable outcomes and exploitation of cell morphology¹¹³.

Although integrating images into deconvolution analyses is very promising, the task is complicated by image data being of a different nature than spatial transcriptomics; morphological differences between cells do not necessarily match transcriptional differences. Additionally, domain shifts between different slides hinder transferability of these methods. Moreover, none of the architectures of methods reviewed here was trained on histopathological slides, which could limit their ability to identify features specific to H&E images. Image-integrated frameworks are also much more computationally demanding than classic frameworks. However, integrating histological features is undoubtedly an important avenue for future developments in this field as it adds morphological information that can help identify cell types.

Incorporating coordinates into deconvolution

Most cell types typically do not localize in isolation but in larger groupings. Spatial transcriptomics data enable the integration of the spatial context, which can help cell-type deconvolution by providing information on spatial proximity¹¹⁴ (Fig. 4c). However, this hypothesis might be violated in certain tissue types with distinct compositions or at the place of histological boundaries, such as highly infiltrated immune cells¹¹⁵, hypoxic environment¹¹⁶ or cancer-associated fibroblasts around tumour nests¹¹⁷.

Several methods incorporate spatial coordinates at various stages of their workflow (Table 1 and Fig. 3). SpatialPrompt³³ and SPADE³⁴ include coordinates during preprocessing of the spatial transcriptomics data by aggregating counts from the 10-nearest neighbours and implementing spatial clustering, respectively. By contrast, other methods use spatial similarity tools such as an adjacency matrix (GraphST⁹¹)

or a smoothing function (BayesTME⁵⁵). Additionally, CARD³⁸ integrates a conditional autoregression model to enforce spatial autocorrelation, whereas SpatialDDL⁸⁵ uses spatial regularization and SONAR³⁶ uses kernel functions to account for distance. Finally, graph structures such as nearest-neighbours graphs and graph convolutional networks are common integration of spatial relationships, such as in GraphST⁹¹. All methods incorporating coordinates typically produce smoother deconvolution results that may be more robust to spatial and technical variation. However, the effectiveness of incorporating spatial coordinates into deconvolution needs further validation through consistent benchmarking and simulations that explicitly incorporate spatial relationships.

Validation and downstream analysis

Given the methodological and conceptual differences in approaches to deconvolution of spatial transcriptomics data, careful validation is essential, albeit challenging, from the perspective of both the developer and the user.

Validating deconvolution results

For developers of deconvolution methods, accuracy and reliability are critical to deconvolution performance. Benchmarking studies have highlighted substantial variation in performance owing to the spatial transcriptomics technology used, number of genes captured and selected, number of spots analysed and number of cell types considered^{101,102,118,119}. Current results show that Cell2location⁴⁸ and RCTD⁵⁴ are top performing deconvolution tools, but the number of benchmarked methods to date is limited and most methods that use deep learning have not yet been benchmarked (Fig. 3 and Supplementary Fig. 2). Additionally, the choice of the benchmarking dataset can influence results^{101,102,118,119}. Strategies to validate deconvolution methods include simulation of pseudo-spots based on scRNA-seq or single-cell resolution (mainly image-based) spatial transcriptomics data and using well-annotated real-world datasets. Beyond simulation, several datasets have been demonstrated as state of the art for validating deconvolution results from spatial transcriptomics data on the mouse brain¹²⁰, human brain¹²¹ and pancreatic carcinoma¹²². These datasets present known histological structure and annotations by pathologists that facilitate method validation.

For users of deconvolution methods, result validation is a major challenge because the ground truth for cell-type composition is often not available. Alternatively, one can apply multiple deconvolution methods and rely on a consensus (based on overlapping results) to increase confidence in the outcomes¹²³. Another approach is to implement ensemble learning, such as in EnDecon¹¹⁸, which combines

Box 2 | Key questions to guide method(s) selection

These 10 questions provide an initial framework for identifying the optimal deconvolution methods for a given objective or project ([web-based summary table](#) and Supplementary Table 2).

What are the tissue characteristics?

It is essential to consider tissue characteristics, including spatial architecture and cell-type composition, because tissues with highly structured organization or defined cell communities may benefit from methods incorporating spatial proximity. By contrast, tissues with variable or disordered spatial patterns may require approaches with more flexibility or no spatial priors to account for their inherent heterogeneity. Moreover, the overlap between cell populations can be assessed from the histopathology image to help guide the choice of method.

How much prior knowledge is available?

Some methods require reference profiles or markers, whereas others are unsupervised. If well-established cell-type profiles are available, reference-based methods may be optimal for high consistency of cellular environments. For limited prior information or to avoid bias between technologies, reference-free methods may be preferable.

Which input datasets are available?

Methods vary based on the input data, which can include single-cell data, spatial coordinates and images (Fig. 3). On the basis of the input data available, the user can narrow down the options also with respect to tissue characteristics and data quality; for example, a low image resolution would disqualify a method using image data.

What is the desired type of output?

Different methods yield varied outputs, some of which might be particularly rare, and the desired type may depend on the question of the project (Fig. 3).

Are model complexity and interpretability crucial?

Deconvolution efforts should begin with simple models to establish feasibility before progressing to more complex approaches to reduce the risk and computational resource waste. Generally, model complexity and interpretability are inversely related. We can rank models (beginning with the least complex) such as: dimensionality reduction, regression, Bayesian, optimal transport and deep learning approaches.

Do the size and dimensionality of the dataset require computational efficiency?

Large or high-dimensional datasets require methods that scale well. Although some approaches are computationally expensive, tools with GPU acceleration may improve scalability (see ref. 101 for benchmarks), which is often reported in the original article.

How much robustness is needed to combat noise and batch effects?

Robust methods are better equipped to handle the noisy signals, batch effects and dropouts that occur with variable data quality. Although often overlooked in spatial transcriptomics, some methods address these issues (see 'Batch_Technology correction' column in the [web-based summary table](#)).

How was the method validated in the original study?

Validation results and benchmarks are reported for each article that presents a new method, in which it is compared with other existing methods. Although these comparisons are not always systematic, they can guide method choice by providing insight into the dataset and benchmark context. Some methods also include tools for internal validation and downstream analysis (see the 'Downstream analysis' column in the [web-based summary table](#)).

Has the method been validated in similar tissue or disease contexts?

Greater confidence can be placed in methods that have been validated in tissues or disease contexts similar to user's samples (see the 'Datasets' column in the [web-based summary table](#)). Moreover, users should explore additional studies driven by biological questions in which the method has been applied and validated to offer further guidance.

How did the method perform in systematic benchmarking (if included)?

Systematic benchmarking studies compare methods in various scenarios using standardized datasets and criteria ([web-based summary table](#) and Supplementary Table 2). Strong performance in such studies enhances confidence in the reliability of a method; however, recent methods may not be benchmarked yet owing to rapid methodological developments.

results from multiple methods using a linear weighted model to provide a consensus along with results from the individual deconvolution algorithms (only from Bayesian and regression frameworks). Another complementary approach is to ensure that the deconvolution results are consistent with prior knowledge, such as the presence of expected cell communities or the expected spatial localization of particular cell types with certain anatomical structures. However, this approach is particularly challenging for complex tissues, such as those found in cancers, that require very-well-annotated pathological features. In such cases, gene signatures from curated databases can be useful for comparing spatial cell-type distribution with signature

scoring, although this solution should be avoided if the deconvolution model already relies on gene signatures. Additionally, the acquisition of imaging-based spatial transcriptomics datasets from the same sample or tissue type can provide a great method of validation. Sequencing-based spatial transcriptomics may help to establish the gene panel, whereas imaging-based spatial transcriptomics achieves single-cell resolution and thus can validate the deconvolution results. Finally, more standard staining methods such as immunostaining can be used to characterize the spatial distribution of cell types with established cell-surface markers, although aligning these different modalities remains challenging.

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Table 1 | Strategies to integrate coordinates and images into deconvolution

Type of integration	Method name ^a	Strategy
Reference-based methods		
Image	GIST	Uses the image data (from immunofluorescence or H&E) with immunofluorescence quantification or deep learning models to generate a prior estimate of the abundance of some cell type at each spot, which is incorporated after a first round of the base model
	<i>RedeHist</i>	Applies U-NET on image patches to extract pixel-level histological features and combines with nucleus masks tensor to generate patch-level features
	Cytospace	Segmentation to estimate cell number
Coordinates	Stereoscope	Location-specific scaling parameter
	SONAR	Weighted kernel function decaying with distance
	SpaTrio	KNN graph
	scDOT	KNN graph
	STEM	Spatial adjacency matrix
	SpatialDDLs	Spatial regularization of proportions by neighbourhood spots
	SD2	Spatial graph, adjacency matrix
	CellTrek	Train multivariate random forest model with [X,Y] ~ PCs
	SpatialPrompt	Weighted mean expression with neighbours' spots
	DOT	Model constraint related to the similarity of expressions in neighbouring spatial locations
	NovoSpaRc	KNN graphs over cells and locations
	STdGCN	Spatial graph
	GNNDeconvolver	Spatial graph, adjacency matrix
	GraphST	KNN graph, adjacency matrix
	EnDecon	Incorporates method-specific constraints
	DEEPsc	Input spatial position in the neural network
	SpaOTsc	Dissimilarity matrix measuring the spatial distance between positions in spatial data
Coordinates and image	SpaDecon	Local smoothing by computing adjacency matrix with respect to spatial locations and histological features
	<i>MAST-Decon</i>	For the coordinates: compute weighted log-likelihood based on location-associated kernel with fixed radius and weight function For the image: image-associated kernel measuring the spot difference in terms of histological features based on embedded feature vectors
	POLARIS	For the coordinates: layer-specific parameter (mean shift in distribution model) For the image: feature extraction with MAE from facebook/MAE for spot and neighbourhood
	Cell2location	For the coordinates: location-specific scaling parameter, hierarchical decomposition prior For the image: segmentation to estimate cell number
	SpatialScope	For the coordinates: Potts model prior with a parameter controlling spatial smoothness of cell-type labels For the image: segmentation to estimate cell number
	Cottrazm	For the coordinates: normalization with SME (spatial location, morphology and gene expression) methodology from stLearn, Manhattan and Euclidean distance For the image: feature extraction with ResNet50/ImageNet, normalization with SME methodology ('morphologically adjusted gene expression matrix')
	SPADE	For the coordinates and image: spatial clustering with SpaGCN
	LETSMix	For the coordinates and image: joint adjacency matrix built on coordinates and histology features
	Tangram	For the coordinates: voxelize the spatial volume at the finest possible resolution and index the voxels in an arbitrary 1D fashion. The main output is a matrix denoting the probability of finding each cell from the scRNA-seq data/snRNA-seq data in each voxel. They propose an optional entropy regularizer to minimize the entropy of the spatial distribution of each cell For the image: segmentation to estimate cell number
Reference-free methods		
Image	scResolve	Pixel-level gene expression inference with Xfuse from the image, cell segmentation
	CellPie	Joint NMF on image and count matrices
	Celloscope	Segmentation to estimate cell number

Table 1 (continued) | Strategies to integrate coordinates and images into deconvolution

Type of integration	Method name ^a	Strategy
Reference-free methods (continued)		
Coordinates	SpiceMix	Probabilistic graph constructed with the neighbours
Coordinates and image	TransformerST	For the coordinates: spot-to-spot correlation graph For the image: feature extraction with a Vision Transformer
Methods with both reference-free and reference-based approaches		
Image	Starfish	A product of experts is used to calculate the joint posterior distribution for gene expression and images
	STIE	Feature extraction from nuclear-morphology-based cell typing
Coordinates	BayesTME	Spatial smoothing hyperparameter
	MUSTANG	Adjacency matrix
	CARD	Conditional autoregressive model

H&E, haematoxylin and eosin; KNN, k-nearest neighbour; MAE, masked autoencoder; NMF, non-negative matrix factorization; PC, principal component; scRNA-seq, single-cell RNA-sequencing; snRNA-seq, single-nucleus RNA-sequencing. ^aItalicized methods are not yet peer-reviewed.

Downstream analysis

Downstream analysis of deconvolution relies on both advanced visualization and statistical techniques to decode the spatial landscape of tissues (Fig. 1d). Spatial statistics has a crucial role in exploring spatial relationships by helping to identify and assess patterns such as cell–cell distances, clustering of cell types and spatial correlation. Methods such as spatially aware clustering and geographically weighted regression enable researchers to detect niche-like environments in which specific combinations of cell types are concentrated^{114,124–126}. These tools can also reveal spatial heterogeneity – that is, variations in the distribution of cell types across different regions of the tissue – providing insights into tissue function and disease progression. However, caution should be taken when performing downstream analyses on cell-type proportions because compositional data require specific treatment (Box 3).

Deconvolution results can be coupled with gene expression imputation to predict cell-specific gene activity and link spatial patterns to potential regulatory mechanisms. Notably, a range of methods have

been developed to infer high-resolution gene expression maps at the pixel level^{127–130}. Furthermore, deconvolution outputs can be analysed with clinical or pathological annotations, such that spatial cell distributions can be connected to clinical outcomes, disease phenotypes and treatment responses¹³¹. Recent advances in cell–cell interaction analysis offer another layer of interpretation¹³². By combining deconvolution data with ligand–receptor interaction networks, one can infer potential communication between neighbouring cells, highlighting the functional dynamics within tissues. This capability is particularly relevant in studies of tumour microenvironments in which complex intercellular interactions drive disease progression.

As the ecosystem of spatial transcriptomics tools continues to grow^{133–135}, downstream analyses of deconvolution results are becoming increasingly flexible to enable the integration of imaging, spatial coordinates and multivariate analysis. These advances will help to refine the understanding of tissue architecture and further our understanding of cell behaviour in space.

Box 3 | Compositional data analysis

Compositional data analysis (CoDA) is a specialized field in statistics and data science focused on analysing data that represent parts of a whole, such as percentages, proportions or concentrations¹⁵⁶. Unlike traditional data analysis, which often assumes absolute values and linear relationships, CoDA deals with data constrained to a constant sum, typically 100% or 1. This constraint introduces specific challenges, such as the need to account for the fact that changes in one component inherently affect others¹⁵⁷. CoDA is crucial in fields such as proteomics, metabolomics and chemistry, in which the relationships between components are as important as their absolute values.

Most deconvolution method outputs are proportions, thus requiring appropriate statistical methods to handle the compositional nature of the data. Traditional methods such as linear regression can lead to misleading results when applied directly to compositional data because they do not account for the relative nature of the data

that breaks the independence assumption of the variables¹⁵⁸. Instead, CoDA uses techniques such as log-ratio transformations to address these issues. For example, the additive log-ratio or centred log-ratio transformations are used to convert compositional data into a format suitable for conventional statistical analyses. These transformations help to mitigate problems associated with the constant sum constraint and enable more accurate modelling of relationships between parts. Moreover, CoDA emphasizes the interpretation of data in terms of relative proportions, which provides insights into the structure of and variation in the composition. This perspective is particularly valuable for understanding complex systems in which interactions between parts have a critical role. By focusing on relative changes rather than absolute amounts, CoDA may offer a nuanced view of deconvolution results, which is essential for making informed decisions and deriving meaningful biological conclusions on cell-type distribution across locations.

Glossary

Archetypal analysis

A statistical technique to identify and summarize patterns in the data by finding archetypes. In deconvolution, these archetypes represent idealized spot composed of a unique cell type. Each spot is then represented as a mixture of these archetypes.

Attention networks

Neural networks equipped with ‘attention mechanisms’ that enable the network to focus on relevant parts of the input data. Attention networks have been used extensively in Natural Language Processing and Computer Vision.

Compositional data

Quantitative descriptions of the parts of some whole, conveying relative information. Common compositional data are proportions and percentages, as encountered in most deconvolution outputs.

Dampened weighted least squares

A classical weighted least-squares regression method with two properties: weights are constrained to be greater than zero and a dampening constant is introduced to prevent infinite weights resulting from low cell-type proportions.

Digital pathology

A branch of pathology that focuses on digitizing and analysing microscopic tissue slides, such as haematoxylin and eosin-stained samples, using high-resolution scanners and advanced artificial intelligence tools.

Dimensionality reduction

A mathematical technique to represent high-dimensional data in lower dimensions. In the context of deconvolution, dimension reduction aims to represent the gene expression data as cell-type contribution in each spot.

Future perspectives

Looking ahead, numerous opportunities exist to improve cell-type deconvolution of spatial transcriptomics and enhance our understanding of how cellular communities are organized and function.

Domain shifts

A phenomenon in machine learning that occurs when training and test data do not follow the same distribution, leading to potential performance degradation. For example, staining protocols might differ between institutes and thus may affect model performance during image analyses.

Ensemble learning

A machine-learning technique consisting in combining multiple models to obtain better predictive performances than individual models.

Foundational models

Large-scale artificial intelligence models trained on broad unlabelled datasets that can be adapted for various downstream tasks.

Fully connected neural network

A neural network in which each neuron in one layer is connected to all neurons in the next layer. Fully connected neural networks are highly flexible, but computationally expensive owing to a large number of parameters. For this reason, neural networks are often only partly fully connected.

Graph neural networks

A type of neural network designed to process data structured as graphs.

Haematoxylin and eosin

(H&E). A gold standard staining used in histology, in which the haematoxylin stains cell nuclei in purplish blue and eosin stains the extracellular matrix and the cytoplasm in pink.

Maximum a posteriori

A probabilistic estimation method that finds the most likely parameter values given the observed data and a prior distribution.

Monte Carlo algorithms

A computational method that relies on random sampling to approximate probability distributions and estimate model parameters. In Bayesian inference, Monte Carlo algorithms — such as Markov chain Monte Carlo — are used to generate samples from complex posterior distributions when direct analytical solutions are intractable.

Multilayer perceptron

A type of fully connected neural network with multiple hidden layers, typically used for classification and regression tasks.

Non-negative least squares

A type of constrained least squares problem in which the coefficients are restricted to be positive.

Optimal transport

A mathematical framework for finding the most efficient way to transform one probability distribution into another while minimizing a specified cost.

Segmentation masks

The result of a segmentation algorithm, which is used to access and further process the objects identified by the segmentation. In the case of tissue images, the segmentation mask can comprise nuclear and cellular regions, or regions of a certain type (for example, tumour or necrotic regions).

Spatial statistics

A field of applied statistics dealing with spatial data leveraging techniques to study entities using their topological, geometric or geographic properties. Spatial transcriptomics data can be analysed with the latest spatial statistics depending on the layout and features of the technology.

Topic modelling

Unsupervised statistical algorithms originally used in text mining for discovering the latent semantic structures of an extensive text body.

Tumour nests

Small clusters or groups of cancerous cells within a tumour, often surrounded by stromal or immune cells. These nests can have a role in tumour growth, invasion and metastasis.

Variational autoencoder

A generative model that learns probabilistic representations of data by mapping inputs into a latent space (encoding). Variational autoencoders are trained with a pretext task, in which a decoder reconstructs the input data from the latent space representation.

Vision transformers

A deep learning model that applies self-attention mechanisms to image patches, representing an alternative to the widely used convolutional neural networks. Vision transformers achieve high performance in computer vision tasks but require large datasets and substantial computational resources owing to the high number of parameters.

Reference-free deconvolution

Reference-free deconvolution frameworks are crucial for analysing uncharacterized or rare tissue types when single-cell data are limited or unavailable. These frameworks excel in handling missing cell types

and mitigating prior annotation biases, which are important functions to potentially uncover novel cell types or states. However, the development of reference-free deconvolution is still limited, particularly in their ability to choose the correct number of cell types and separate the signals from transcriptionally close cell types. Furthermore, the lack of a common annotation pipeline across methodologies may lead to challenges in consistently annotating the output.

Robust deconvolution for comparative analyses

As the popularity of spatial transcriptomics grows, flexible deconvolution techniques capable of multi-sample and multi-condition analyses are becoming increasingly important. New methodological development is also needed to handle the different experimental designs across datasets and atlases, as few methods currently control for technology effects. Moreover, the propensity of spatial data to suffer from batch effects necessitates integration methods tailored specifically to account for the spatial dimension^{136–138}, of which only a few methods currently implement ([web-based summary table](#) and Supplementary Table 2). To enable meaningful comparative analyses – such as assessing cell-type enrichment across multiple slides, locations and conditions – it is essential that deconvolution methods provide consistent and comparable results across spatial transcriptomics datasets. Comparative analyses enabled by such robust methods have the potential to uncover richer spatial patterns and improve our understanding of tissue organization.

Multimodal deconvolution

Incorporating scRNA-seq data, images and spatial coordinates into deconvolution workflows can substantially enhance analyses of spatial transcriptomics data by improving the resolution and accuracy of cell-type mapping. However, challenges remain in integrating multiple modalities, including building comprehensive reference datasets and controlling for technology effects, which must be carefully managed through rigorous validation. Developing deconvolution tools that integrate images is an especially major future challenge. Digital pathology already uses complex image-based models effectively, suggesting that similar techniques could advance spatial transcriptomics deconvolution. As such, high-quality scans that accompany spatial transcriptomics could enhance feature extraction, model accuracy and gene expression imputation. Additionally, combining spatial transcriptomics with other modalities^{13,139–141}, such as protein and chromatin accessibility information, could provide valuable insights. Finally, cell surface markers have been extensively used in cytometry and immunostaining to characterize cell types and thus could serve as both an additional layer of data and a validation tool for multimodal spatial deconvolution^{142,143}.

Comprehensive benchmarking

The sheer number of deconvolution methods has made comprehensive benchmarking impractical, and validating the results from these methods is challenging, particularly for tissues with no distinct spatial architecture. Indeed, only some methods have been assessed in systematic benchmarks (Supplementary Fig. 2). To address these issues, novel approaches are needed for effective benchmarking and result validation, such as those that evaluate the impact of integrating spatial coordinates and images on performance, as more methods now incorporate these modalities. The release of curated, diverse real-world datasets for benchmarking is essential, as current datasets mostly focus on common tissues such as mouse and human brains. To motivate fair and comprehensive benchmarking, dedicated challenges

could be organized, as in many computational fields^{20,144,145}. Moreover, the integration of image and omics data will require pathologists to be involved in the data analysis and development of cross-validation methods for new tissues.

Deconvolution at increasing resolution

Spatial transcriptomics is rapidly advancing towards high-throughput, spatial, single-cell transcriptomics with developments in both sequencing-based and imaging-based approaches. Visium HD and Slide-seqV2 (ref. 146) are pushing the boundaries of resolution, whereas MERFISH¹⁴⁷ and Xenium¹⁴⁸ are scaling up to thousands of genes. As a result, the growing repository of publicly available spatial transcriptomics^{113,149–154} is becoming an invaluable resource across a wide range of research areas, enabling large-scale, integrative analyses. In this context, the re-analysis of large cohorts of spatial transcriptomics data generated with low-resolution technologies highlights the ongoing need for deconvolution methods. These methods may also provide complementary insights alongside higher-resolution spatial technologies, such as the use of bulk deconvolution with single-cell data. As such, the development of robust integration methods remains crucial to fully harness the spatial and molecular complexity revealed by high-resolution platforms.

As the field advances, emerging technologies will demand not only deconvolution techniques but also innovative approaches to handle data aggregation. For example, Visium HD now has a 2 μm^2 (subcellular) resolution, but the barcoding area is organized as a grid that hinders precise capture of RNA within individual cells. This limitation will require new methods to aggregate capture areas and reconstruct single-cell transcriptomes within cell boundaries¹⁵⁵, which will likely require a subsequent deconvolution step to identify cell types. For imaging-based technologies, single-cell transcriptomes will become more precise with advances in cell segmentation and throughput, leading to the need for methods that leverage label transfer from scRNA-seq data and better integrate staining and spatial information. These technological and methodological improvements will help to ultimately predict gene expression and spatial patterns across varying scales and resolutions, enabling the reconstruction and modelling of cellular environments. The mathematical foundations established by deconvolution methods will be invaluable for the continuous integration of image, spatial coordinates and transcriptomics data. Together, these advancements will propel spatial transcriptomics into new realms of discovery, enhancing our understanding of cellular function and spatial organization within tissues.

Conclusion

Cell-type deconvolution is undergoing incredible advances, providing new insights into the organization of cellular communities from spatial transcriptomics. This progress holds tremendous potential for future applications such as dissecting developmental processes, deciphering tumour microenvironment architecture and understanding virus–host interactions in tissues. Methods for cell-type deconvolution inference vary in their strengths, assumptions and limitations, which are crucial to understand when applying these strategies and selecting the most suitable parameters for analysis. This active research field holds tremendous potential to enhance our understanding of cellular organization and functions of complex systems as revealed by spatial transcriptomics.

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Author contributions

L.C.G.-B. and L.G. researched the literature. All authors contributed substantially to discussion of the content. L.C.G.-B. and L.G. wrote the article. All authors reviewed and edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

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